

Package #169

Incorporate a marine-grade polymer anti-biofouling coating (e.g., copper-based or silicone-based fouling-release polymer) onto catheter surfaces to create a non-thrombogenic interface.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'catheter occlusion' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1997-2032). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000032: Malfunction [PUMP MMT-1780KPK 670G PATHWAY BLACK MG]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'The Challenges of Vascular Implants: Regulatory Strategies and Biological Responses.'; 'Emerging Gel Technologies for Atherosclerosis Research and Intervention.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Antifouling surface coating that chemically resists adhesion of biological or material deposits, as evidenced by heparin coating's anticoagulant effect.

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.